Mateo Lopez Elizondo

Stanford, CA · matlo202@stanford.edu · linkedin.com/in/mateolopezelizondo · github.com/matmat202

EDUCATION

Stanford University, Stanford, CA — *B.S. candidate, Mathematics and Management Science & Engineering (intended)* · Expected Jun 2029

GPA: 3.76/4.00

Relevant coursework: CS 205L Continuous Mathematical Methods for Machine Learning; MATH 151 Introduction to Probability Theory; CS 106B Programming Abstractions; MATH 63DM Modern Mathematics: Discrete Methods; STATS 200 Introduction to Theoretical Statistics, MS&E 319 Efficient Generative Models, MATH 175 Elementary Functional Analysis, EE 102A Signals and Systems I (in progress).

EXPERIENCE

Alvarez & Marsal — Consultant Intern, Performance Improvement · Mexico City, Mexico · Jun 2026 – Sep 2026

- Automated ingestion and normalization of thousands of heterogeneous shipment files across 20+ carriers in Python, reducing multi-day workflows to minutes; drafted the carrier-mapping logic with an AI assistant, then wrote validation checks reconciling its output against source totals.
- Built a Python transportation-optimization platform on that data, modeling FTL/LTL selection, capacity, delivery windows, driving-hour rules, carrier pricing and routing constraints.
- Backtested optimized routes over 26+ weeks of historical operations; modeled an 11.3% reduction in FTL routing cost under matched-rate assumptions, about $1.2M in projected annualized savings.
- Analyzed 500K+ rows across 9,000+ locations for a separate engagement; selected to lead a one-hour Python and AI automation workshop for 15–20 colleagues.

Red-Horse Lab, Stanford University — Research Assistant · Stanford, CA · Mar 2026 – Jun 2026

- Developed a 3D vessel-filling algorithm reconstructing vessel interiors from noisy boundary data with physics-based balloon models, data cleansing and convergence control, for research on arterial tortuosity, volume and cross section.

Blyth Fund at Stanford — Portfolio Intern · Stanford, CA · Jan 2026 – Jun 2026

- Supported management of a $371K student-run long-only equity portfolio; researched companies, industries and valuations, and evaluated student stock pitches.

Viva Aerobus — Networking Strategy Intern · Monterrey, Mexico · Jul 2023 – Dec 2023

- Proposed and then led demand-forecasting work in PyTorch to anticipate competitor route openings, including game-theoretic competitive dynamics; proposed data-collection pipelines and visualized market flow and model decisions over time for the network team.

PROJECTS

Stanford ACM — Quant Gym, MLab, DevLab · Sep 2025 – Present

- Quant Gym (probability, statistics, modeling); MLab (machine-learning architectures; research on language-model training and evaluation); DevLab (full-stack Next.js applications).

C++ Particle Simulation · Personal project · Oct 2025

- Wrote a collision simulator in C++ handling thousands of vectorized particles and multiple sources, modeling fireworks, black holes, electric fields and particle-based images.

International Economics Olympiad, Team Mexico — Gold Medalist · Apr 2025 – Jul 2025

- Led the quantitative modeling for the Business Case: piecewise cost functions for bridge construction against water depth, and A*/Dijkstra pathfinding over bathymetric grid data; Mexico placed 2nd of 54 countries, 11th worldwide individually.

LEADERSHIP

Stanford Latino — Director of Internships (Jun 2026 – Present), Board Member (Sep 2025 – May 2026) · **SOLE** — Academic Officer (Mar 2026 – Present); rebuilt a legacy contacts system, 5x usable contacts · **ASFM Mexican Math Olympiad Team** — Founder (Apr 2024 – Jun 2025): started the school's first olympiad team and brokered a training system with state organizers.

OMM Silver Medalist (2024) · ASFM Valedictorian (2025).

SKILLS

Python, C++, Java, SQL, JavaScript, LaTeX · PyTorch, Next.js, Git, Linux · optimization, mathematical modeling, simulation, forecasting, data visualization · English and Spanish (fluent)